Duty at home

DM2 · Approaching a root · 50 min.

It is considered $u_0 = 0$ and, for all $n \in \mathbb{N}$, $u_{n+1} = \sqrt{2 + u_n}$. Questions 1, 2 and 4 are to be dealt with after S20; question 3 will be resumed after the course on continuity in S31.

1. Demonstrate by induction that all terms are defined and that $0 \leq u_n \leq 2$ for all $n \in \mathbb{N}$.

2. Demonstrate by induction that $u_{n+1} \geq u_n$ for every $n \in \mathbb{N}$. Deduce that (u_n) converge.

3. Using the continuity of the function $x \mapsto \sqrt{2 + x}$, determine the limit of (u_n).

4. Demonstrate $0 \leq 2 - u_{n+1} \leq \frac{2 - u_n}{2}$ for every $n \in \mathbb{N}$. Deduct an increase of $2 - u_n$ in function of n, then one integer N such as $0 \leq 2 - u_n \leq 10^{-3}$ for every $n \geq N$.